

HW # 34

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CS-F

Prob 1

$$0 = m + c$$

$$1 = 2m + c$$

$$3 = 3m + c$$

$$\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ m \end{bmatrix}$$

$$b = Ax$$

~~$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & -1 \\ 1 & 2 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 3 & 3 & 1 & 3 & 3 & 1 & 3 & 3 \end{bmatrix}$$~~

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

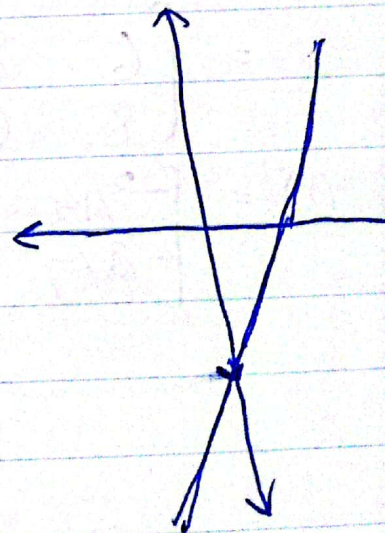
$$A^T A = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$(A^T A)^{-1} = \begin{bmatrix} 7/3 & -1 \\ -1 & 1/2 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 11 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} -5/3 \\ 3/2 \end{bmatrix}$$

$$y = \frac{3}{2}x - \frac{5}{3}$$



Prob 2

$$(-1, 8) \quad (0, 8) \quad (1, 4) \quad (2, 16)$$

$$8 = c_0 + (-1)c_1 + (-1)^2 c_2$$

$$c_0 = 8 \rightarrow \text{(i)}$$

$$8 = c_0 - c_1 + c_2 \rightarrow \text{(i)}$$

$$4 = c_0 + c_1 + c_2 \rightarrow \text{(ii)}$$

$$16 = c_0 + 2c_1 + 4c_2 \rightarrow \text{(iv)}$$

$$0 = -c_1 + c_2$$

$$-4 = c_1 + c_2$$

$$8 = 2c_1 + 4c_2 \Rightarrow 4 = c_1 + 2c_2$$

$$\begin{bmatrix} 0 \\ -4 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$ATA = \begin{bmatrix} -1 & 1 & 2 \\ 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 8 \\ 8 & 6 \end{bmatrix}$$

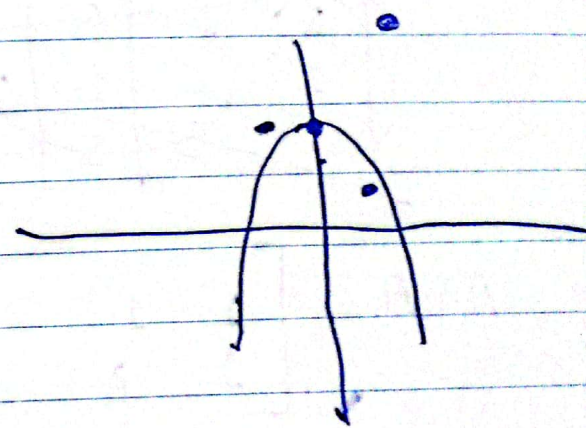
$$(ATA)^T = \begin{bmatrix} -3/14 & -2/7 \\ -2/7 & -3/14 \end{bmatrix}$$

$$ATb = \begin{bmatrix} -1 & 1 & 2 \\ 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ -4 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} -6/7 \\ -8/7 \end{bmatrix}$$

$$y = 8 - \frac{6}{7}x - \frac{8}{7}x^2$$



Prob 3

$$(1, 3) \quad (-1, 13) \quad (2, 1) \quad (-2, 33)$$

$$y = c_0 + c_1x + c_2x^2 + c_3x^3$$

$$3 = c_0 + c_1 + c_2 + c_3$$

$$13 = c_0 - c_1 + c_2 - c_3$$

$$1 = c_0 + 2c_1 + 4c_2 + 8c_3$$

$$33 = c_0 - 2c_1 + 4c_2 - 8c_3$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 2 & 4 & 8 \\ 1 & -2 & 4 & -8 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 13 \\ 1 \\ 33 \end{bmatrix}$$

A X b

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 2 & -2 \\ 1 & 1 & 4 & 4 \\ 1 & -1 & 8 & -8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 2 & 4 & 8 \\ 1 & -2 & 4 & -8 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 & 10 & 0 \\ 0 & 10 & 0 & 34 \\ 10 & 0 & 34 & 0 \\ 0 & 34 & 0 & 130 \end{bmatrix}$$

~~$(A^T A)^{-1}$~~ $\rightarrow$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 2 & -2 \\ 1 & 1 & 4 & 4 \\ 1 & -1 & 8 & -8 \end{bmatrix} \begin{bmatrix} 3 \\ 13 \\ 1 \\ 33 \end{bmatrix}$$

$$= \begin{bmatrix} 50 \\ -74 \\ 152 \\ -266 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 & 10 & 0 & 50 \\ 0 & 10 & 0 & 34 & -74 \\ 10 & 0 & 34 & 0 & 152 \\ 0 & 34 & 0 & 130 & -266 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

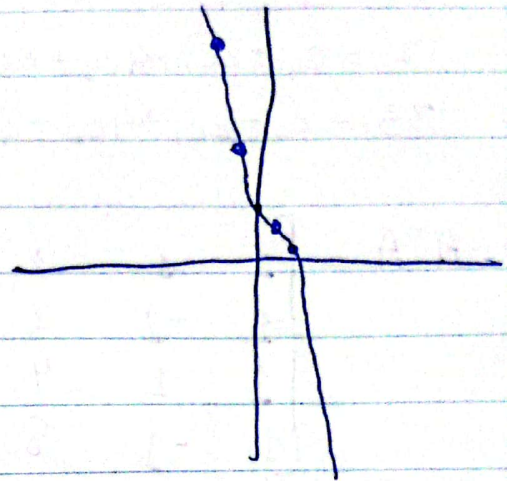
$$c_0 = 5$$

$$c_1 = -4$$

$$c_2 = 3$$

$$c_4 = -1$$

$$y = 5 - 4x + 3x^2 - x^3$$



Prob 4

$$(8, 1) \quad (0, 2) \quad (10, 4) \quad (10, 5) \quad (2, 6)$$

$$\begin{aligned} 1 &= 8m + c \\ 2 &= c \\ 4 &= 10m + c \\ 5 &= 10m + c \\ 6 &= 2m + c \end{aligned}$$

$$\begin{bmatrix} 1 \\ 2 \\ 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 & 8 \\ 1 & 0 \\ 1 & 10 \\ 1 & 10 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix}$$

b A

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 8 & 0 & 10 & 10 & 2 \end{bmatrix} \begin{bmatrix} 1 & 8 \\ 1 & 0 \\ 1 & 10 \\ 1 & 10 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 30 \\ 30 & 268 \end{bmatrix}$$

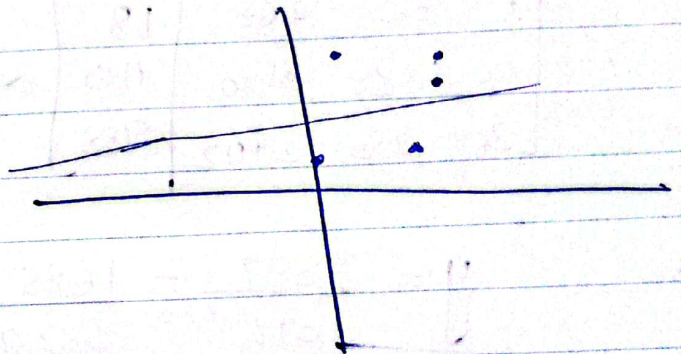
$$(A^T A)^{-1} = \begin{bmatrix} 67/110 & -3/44 \\ -3/44 & 1/88 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 8 & 0 & 10 & 10 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 18 \\ 110 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 67/110 & -3/44 \\ -3/44 & 1/88 \end{bmatrix} \begin{bmatrix} 18 \\ 110 \end{bmatrix}$$

$$= \begin{bmatrix} 381/110 \\ 1/44 \end{bmatrix}$$



$$y = \frac{1}{44} x + \frac{381}{110}$$

$$1 = c_0 + 8c_1 + 64c_2$$

$$2 = c_0$$

$$4 = c_0 + 10c_1 + 100c_2$$

$$5 = c_0 + 10c_1 + 100c_2$$

$$6 = c_0 + 2c_1 + 4c_2$$

$$\begin{bmatrix} 1 \\ 2 \\ 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 & 8 & 64 \\ 1 & 0 & 0 \\ 1 & 10 & 100 \\ 1 & 10 & 100 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 8 & 0 & 10 & 10 & 2 \\ 64 & 0 & 100 & 100 & 4 \end{bmatrix} \begin{bmatrix} 1 & 8 & 64 \\ 1 & 0 & 0 \\ 1 & 10 & 100 \\ 1 & 10 & 100 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 30 & 268 \\ 30 & 268 & 2520 \\ 268 & 2520 & 24112 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 8 & 0 & 10 & 10 & 2 \\ 64 & 0 & 100 & 100 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 18 \\ 110 \\ 988 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 5 & 30 & 268 & 18 \\ 30 & 268 & 2520 & 110 \\ 268 & 2520 & 24112 & 988 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3887/1016 \\ 0 & 1 & 0 & -1685/4064 \\ 0 & 0 & 1 & 343/8128 \end{array} \right]$$

$$y = \frac{3887}{1016} - \frac{1685}{4064}x + \frac{343}{8128}x^2$$

